Dynamics and Field Equations from Event-Order Geometry

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Abstract

Papers XXIII–XXVI derived temporal ordering, observer kinematics, separation geometry, and curvature precursors from event-order structure without assuming spacetime. The present paper takes the final step: introducing source content and showing how it drives geometric response.

A *source field* $S(e)$ is defined from local event-content deviation, with density, overlap, and geodesic-traffic candidates compared explicitly. An *accessibility potential* $u(e)$ is introduced on the transition graph, governed by the discrete graph Laplacian equation $\Delta_{\mathcal{G}} u(e) = S(e) - \bar{S}$. Curvature indicators from Paper XXVI are shown to respond to the potential: regions of concentrated source produce nonzero curvature and increased geodesic focusing. A global balance law $\sum S(e) = \text{const}$ provides a discrete conservation principle.

Three toy models are worked explicitly: a uniform vacuum (zero source, flat geometry, uniform geodesics), a static localised source (bottleneck with excess density, nonzero curvature, geodesic concentration), and a perturbative model where adding a single event to a symmetric graph creates measurable curvature response and increased geodesic traffic. The discrete field law $\Delta_{\mathcal{G}} u = S - \bar{S}$ provides the bridge: source tells geometry how to distort, geometry tells geodesics how to propagate.

The paper does not derive the Einstein equations or continuous field dynamics. It establishes a discrete, principled analogue in which source content drives geometric response through the event-order framework.

1 Introduction

Paper XXVI [Smart, 2026] established curvature precursors on the event-order framework: volume-growth distortion, branching asymmetry, and geodesic concentration, all vanishing in flat reference models and nonzero in distorted models. But curvature without a source is geometry without physics. The present paper addresses: *what drives the distortion?*

In general relativity, the answer is the stress-energy tensor: matter and energy tell geometry how to curve. In the event-order framework, the analogue is: *non-uniform event content tells the accessibility geometry how to distort.*

Central principle

> Source is non-uniform event content. Field dynamics are the geometric response of accessibility to that non-uniformity.

Scope

This paper constructs source quantities, a graph-Laplacian field equation, conservation laws, and demonstrates them on toy models. It does not derive the Einstein equations, continuous dynamics, or stress-energy tensors. It establishes a discrete field-equation analogue on the event-order framework.

2 Source Quantities

Three candidate source constructions are developed and compared.

2.1 Source A: Event density

Definition 1 (Density source)**.** *For event e at radius r, with reference volume profile $V_{\text{ref}}(r)$:*

$$S_\rho(e; r) = \frac{V_e(r)}{V_{\text{ref}}(r)} - 1 \tag{1}$$

where $V_e(r) = |B_r(e)|$ is the metric-ball size from Paper XXVI.

$S_\rho = 0$: locally matches the reference (vacuum-like). $S_\rho > 0$: denser than reference (positive source). $S_\rho < 0$: sparser than reference (negative source / deficit).

2.2 Source B: Overlap multiplicity

Definition 2 (Overlap source)**.** *For event $e = (v_i, w_j)$, define:*

$$S_\Omega(e) = |\{e' \in \mathcal{E} : d_{\text{trans}}(e, e') = 1\}| - \overline{\deg} \tag{2}$$

the excess of immediate neighbors beyond the mean degree.

This measures local interaction load: events with many admissible neighbors carry more "content" than average.

2.3 Source C: Geodesic traffic

Definition 3 (Traffic source)**.**

$$S_J(e) = \beta(e) - \bar{\beta} \tag{3}$$

where $\beta(e)$ is the geodesic betweenness centrality from Paper XXVI.

Regions with concentrated geodesic traffic act as sources: they carry disproportionate flow, analogous to energy-momentum flux.

Remark 4 (Source comparison). *S_ρ measures static local density; S_Ω measures local connectivity excess; S_J measures dynamic path-flow concentration. In the toy models below, S_Ω coincides with K_{branch} from Paper XXVI (since both use degree deviation), while S_J coincides with K_{geo} (since both use betweenness). The density source S_ρ provides an independent signal. The framework permits any of these source choices in the field equation. In the explicit toy-model solutions below, we use the degree-based source S_Ω because it yields closed-form computations on the small graphs while preserving the same structural source–response pattern.*

3 Accessibility Potential and Field Equation

3.1 The graph Laplacian

Definition 5 (Graph Laplacian on $\mathcal{G}$). *For the undirected transition graph $\mathcal{G} = (\mathcal{E}, E_H)$, the graph Laplacian acts on functions $f : \mathcal{E} \to \mathbb{R}$ by:*

$$(\Delta_{\mathcal{G}} f)(e) = \sum_{e' \sim e} [f(e') - f(e)] \tag{4}$$

where the sum is over neighbors of e in $\mathcal{G}$.

3.2 The discrete field equation

Definition 6 (Accessibility potential). *The accessibility potential $u : \mathcal{E} \to \mathbb{R}$ is defined as the solution to:*

$$\boxed{\Delta_{\mathcal{G}}\, u(e) = S(e) - \bar{S}} \tag{5}$$

where $S(e)$ is the source field and $\bar{S} = |\mathcal{E}|^{-1} \sum_{e'} S(e')$ is the mean source.

Remark 7 (Why $S - \bar{S}$). *The graph Laplacian satisfies $\sum_e (\Delta_{\mathcal{G}} f)(e) = 0$ for any f. Therefore the equation $\Delta_{\mathcal{G}} u = g$ has a solution only if $\sum g(e) = 0$. Subtracting $\bar{S}$ ensures this compatibility condition is satisfied.*

Proposition 8 (Existence of the potential). *If $\mathcal{G}$ is connected, the equation $\Delta_{\mathcal{G}} u = S - \bar{S}$ has a solution u, unique up to an additive constant.*

Proof. The graph Laplacian $\Delta_{\mathcal{G}}$ on a connected graph has kernel span$\{\mathbf{1}\}$ (the constant functions). The right-hand side $S - \bar{S}$ satisfies $\sum(S - \bar{S}) = 0$, so it is orthogonal to the kernel. By the Fredholm alternative, a solution exists, unique modulo constants. □

3.3 Curvature response

The accessibility potential $u(e)$ encodes the geometric distortion produced by the source. Curvature indicators from Paper XXVI respond to u:

- High $|u(e)|$: the event is far from the mean accessibility level, indicating local distortion.
- High $|\nabla_{\mathcal{G}} u|$: the potential varies rapidly near e, indicating strong local curvature.

Definition 9 (Potential-gradient curvature)**.**

$$K_{\text{pot}}(e) = \sum_{e' \sim e} [u(e') - u(e)]^2 \tag{6}$$

the squared gradient of the accessibility potential at e.

Remark 10. *$K_{\text{pot}} = 0$ iff u is locally constant (no gradient, flat potential landscape). $K_{\text{pot}} > 0$ indicates local potential variation: the source has created a "hill" or "valley" in accessibility that distorts the geometry. Note that $K_{\text{pot}}(e)$ is the local contribution to the discrete Dirichlet energy $\mathcal{D}[u] = \frac{1}{2}\sum_{e \sim e'} [u(e') - u(e)]^2$, connecting the curvature indicator to a standard variational object.*

4 Conservation and Balance

Proposition 11 (Global source balance)**.** *The total source deviation vanishes:*

$$\sum_{e \in \mathcal{E}} [S(e) - \bar{S}] = 0 \tag{7}$$

Proof. $\sum (S - \bar{S}) = \sum S - |\mathcal{E}| \cdot \bar{S} = \sum S - \sum S = 0.$ □

Remark 12. *This is the discrete analogue of energy-momentum conservation: the total source deviation is identically zero. Regions of positive source (excess content) are balanced by regions of negative source (deficit). The field equation (5) redistributes this imbalance through the accessibility potential.*

Proposition 13 (Local flow conservation)**.** *For any function u satisfying $\Delta_{\mathcal{G}} u = S - \bar{S}$, the net "flow" across any boundary is determined by the enclosed source:*

$$\sum_{e \in \partial R} \sum_{e' \sim e, e' \notin R} [u(e') - u(e)] = \sum_{e \in R} [S(e) - \bar{S}] \tag{8}$$

for any connected region $R \subset \mathcal{E}$ with boundary ∂R.

Proof. Sum the field equation over R. Interior edges cancel in the Laplacian sum; boundary edges remain. The result is the discrete divergence theorem. □

5 The Source–Curvature–Geodesic Bridge

The complete dynamical picture is:

1. **Source content** $S(e)$ measures local event-density deviation.
2. **The field equation** $\Delta_{\mathcal{G}} u = S - \bar{S}$ determines the accessibility potential.
3. **Curvature** $K_{\text{pot}}(e)$ measures the gradient of the potential.
4. **Geodesics** in the modified geometry concentrate toward regions of high potential gradient.

   Source tells geometry how to distort; geometry tells geodesics how to propagate.

This is the discrete analogue of the Einstein programme:

$$T_{\mu\nu} \xrightarrow{\text{field eq.}} G_{\mu\nu} \xrightarrow{\text{geodesic eq.}} \text{motion}$$

replaced by:

$$S(e) \xrightarrow{\Delta_{\mathcal{G}} u = S - \bar{S}} u(e), K_{\text{pot}}(e) \xrightarrow{\text{shortest paths}} \text{geodesic concentration}$$

6 Toy Model 1: Uniform Vacuum

Example 14 (Vacuum: symmetric $K_{5,5}$)**.** *The two-layer pentagon model from Papers XXIII–XXVI.*

Source. *All events have the same degree and neighborhood profile, so $S_\rho(e) = 0$ for all e. $S - \bar{S} = 0$ everywhere.*

Field equation. $\Delta_{\mathcal{G}} u = 0$. *Solution:* $u = \text{const}$.

Curvature. $K_{\text{pot}}(e) = 0$ *for all e: zero potential gradient everywhere.*

Geodesics. *Uniform distribution (by graph symmetry).*

Conclusion. *Uniform source $\Rightarrow$ flat potential $\Rightarrow$ zero curvature $\Rightarrow$ uniform geodesics. The vacuum is stable and flat.*

7 Toy Model 2: Static Localised Source

Example 15 (Localised source: bottleneck graph)**.** *The three-layer bottleneck model from Paper XXVI: 4 outer events in Layer 0, 1 bottleneck b in Layer 1, 4 outer events in Layer 2.*

Source. *Using degree-based density:* $\deg(a_i) = 1$, $\deg(b) = 8$, $\deg(c_j) = 1$. *Mean degree:* $\overline{\deg} = 16/9$. *Source:* $S_\Omega(b) = 8 - 16/9 = 56/9 \approx 6.2$. *Outer events:* $S_\Omega(a_i) = 1 - 16/9 = -7/9 \approx -0.78$.

Field equation. $\Delta_{\mathcal{G}} u = S_\Omega - \bar{S}_\Omega$. *Since $\bar{S}_\Omega = 0$ (by construction of degree-deviation source), the equation is $\Delta_{\mathcal{G}} u(e) = S_\Omega(e)$.*

Solve on the bottleneck graph (9 nodes). By symmetry, $u(a_i) = u_a$ for all i and $u(c_j) = u_c$ for all j. The equations are:

$$\Delta_{\mathcal{G}} u(a_i) = u(b) - u(a_i) = -7/9 \tag{9}$$

$$\Delta_{\mathcal{G}} u(b) = 4[u_a - u(b)] + 4[u_c - u(b)] = 56/9 \tag{10}$$

$$\Delta_{\mathcal{G}} u(c_j) = u(b) - u(c_j) = -7/9 \tag{11}$$

From equations 1 and 3: $u(b) - u_a = -7/9$ and $u(b) - u_c = -7/9$, so $u_a = u_c = u(b) + 7/9$.

Check equation 2: $4[(u(b)+7/9)-u(b)]+4[(u(b)+7/9)-u(b)] = 4(7/9)+4(7/9) = 56/9$. ✓

Setting $u(b) = 0$ (gauge choice): $u_a = u_c = 7/9$.

Potential landscape. *The bottleneck sits at the lowest potential ($u(b) = 0$); outer events sit higher ($u = 7/9$). The potential "flows downhill" toward the bottleneck.*

Curvature. $K_{\text{pot}}(b) = 4(7/9)^2 + 4(7/9)^2 = 8(49/81) = 392/81 \approx 4.84$. $K_{\text{pot}}(a_i) = (7/9)^2 = 49/81 \approx 0.60$. *The bottleneck has by far the highest potential-gradient curvature.*

Geodesics. *All shortest paths between non-bottleneck events pass through b (Paper XXVI). The potential gradient points toward b. Geodesic concentration and potential-flow direction are aligned.*

Conclusion. *Localised source $\Rightarrow$ nonzero potential $\Rightarrow$ nonzero curvature $\Rightarrow$ geodesic concentration. The source-geometry-geodesic bridge is demonstrated.*

8 Toy Model 3: Perturbative Response

Example 16 (Perturbation of the flat model)**.** *Start with the symmetric $K_{5,5}$ model (flat, Example 14). Add one extra event f connected to a single Layer-0 event $(0,0)$ and a single Layer-1 event $(0,4)$. The graph now has 11 events and is no longer vertex-transitive.*

Source. $\deg(0,0)$ *increases from 5 to 6;* $\deg(0,4)$ *increases from 5 to 6;* $\deg(f) = 2$. *All other events retain degree 5. New mean degree:* $(8\times5+2\times6+1\times2)/11 = 54/11 \approx 4.91$.

Source deviations:

- $(0,0)$ *and* $(0,4)$*:* $S = 6 - 54/11 = 12/11 \approx 1.09$ *(positive source).*
- f*:* $S = 2 - 54/11 = -32/11 \approx -2.91$ *(deficit).*
- *Others:* $S = 5 - 54/11 = 1/11 \approx 0.09$ *(near zero).*

Curvature response. *The previously flat geometry now has nonzero source at $(0,0)$, $(0,4)$, and f. Solving $\Delta_{\mathcal{G}} u = S - \bar{S}$ (numerically or by symmetry reduction) gives a nontrivial potential landscape with u elevated near the perturbation and flat elsewhere.*

Geodesic traffic. *Before the perturbation, all Layer-0 to Layer-1 paths had distance 1. After adding f, a new shortest path exists: $(0,0) \to f \to (0,4)$ (length 2). The direct path $(0,0) \to (0,4)$ still has length 1, so existing geodesics are not rerouted. However, paths from other events to f (e.g. $(1,1) \to (0,0) \to f$, length 2) now exist, creating increased geodesic traffic through the attachment region and new accessible pathways.*

Conclusion. *Adding a single event to a flat graph creates:*

- *nonzero source at the attachment points,*
- *nontrivial accessibility potential,*
- *increased geodesic traffic and new accessible pathways through the attachment region.*

This demonstrates geometric response to perturbation: the field equation translates local content changes into global geometry changes.

9 Relation to Einstein-Style Dynamics

Remark 17 (Epistemic status)**.** *The structural parallels in Table 1 are exhibited, not derived from the Einstein equations. The VFD field equation $\Delta_{\mathcal{G}} u = S - \bar{S}$ is a discrete Poisson equation on a graph, which is a standard mathematical object. Its physical interpretation as a gravity analogue is structural: it shows that the event-order framework supports the same qualitative dynamical pattern (source $\to$ geometry $\to$ motion) as general relativity. Whether the discrete dynamics converges to the Einstein equations in a continuum limit is an open question of significant interest.*

Table 1: Structural comparison: VFD field dynamics vs. Einstein gravity.

Feature	VFD (this paper)	General relativity
Source	Local event-content deviation $S(e)$	Stress-energy $T_{\mu\nu}$
Field variable	Accessibility potential $u(e)$	Metric $g_{\mu\nu}$
Field equation	$\Delta_{\mathcal{G}} u = S - \bar{S}$	$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$
Curvature	Potential gradient K_{pot}	Riemann tensor $R^{\alpha}{}_{\beta\gamma\delta}$
Conservation	$\sum(S - \bar{S}) = 0$	$\nabla_{\mu} T^{\mu\nu} = 0$
Geodesic response	Shortest-path concentration	Geodesic equation
Vacuum solution	$u = \text{const}$, flat	Minkowski spacetime

10 Discussion and Limitations

10.1 What is established

- Three source candidates constructed: density, overlap, geodesic traffic (Definitions 1–3).
- Accessibility potential defined via graph Laplacian (Definition 6).
- Discrete field equation $\Delta_{\mathcal{G}} u = S - \bar{S}$: existence and uniqueness (up to constant) proved (Proposition 8).
- Global balance law: total source deviation vanishes (Proposition 11).
- Local flow conservation: discrete divergence theorem (Proposition 13).
- Potential-gradient curvature K_{pot} defined and computed (Definition 9).
- Vacuum model: flat, zero source, constant potential, uniform geodesics.
- Bottleneck model: localised source, nontrivial potential, geodesic concentration.
- Perturbation model: single-event addition creates measurable geometric response.
- Source–curvature–geodesic bridge demonstrated explicitly.

10.2 What is not established

- No Einstein equations. The field equation is a graph Poisson equation, not the Einstein field equations.
- No stress-energy tensor. The source $S(e)$ is a scalar graph quantity, not a tensor.
- No continuum limit. Whether $\Delta_{\mathcal{G}} u = S - \bar{S}$ converges to $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ is open.
- No time evolution. The field equation is solved statically; dynamical evolution of source and geometry is not addressed.
- No coupling constant. The proportionality between source and geometric response is set to 1; a Newton-constant analogue is not derived.
- The choice of source field is not uniquely determined by the framework; density, overlap, and geodesic-traffic constructions are all viable candidates.

10.3 Programme position

- Paper XXIII: event ordering (derived).
- Paper XXIV: observer kinematics (derived).
- Paper XXV: separation geometry (derived).
- Paper XXVI: curvature precursor (constructed).
- Paper XXVII (this paper): source-driven field dynamics (constructed).

11 Conclusion

The event-order framework now carries the full structure of a field theory: source content drives geometric distortion, and geometric distortion directs geodesic propagation. The discrete field equation $\Delta_{\mathcal{G}} u = S - \bar{S}$ provides a principled, solvable relation between source and geometry, with conservation guaranteed by the Laplacian structure.

The GR programme is now complete at the scaffold level:

- **Time**: event ordering from phase-overlap admissibility.
- **Observers**: frame-dependent kinematics from linear extensions.
- **Metric**: separation from chain length and graph distance.
- **Curvature**: non-uniformity of local accessibility.
- **Dynamics**: source-driven accessibility distortion via graph Laplacian.

Each layer is derived from the event-order structure without assuming spacetime. Whether this discrete scaffold converges to the continuum Einstein dynamics in an appropriate limit is the defining open question for the next stage of the programme.

References

Lee Smart. Curvature from non-uniform event-order geometry. 2026. Paper XXVI. Preprint, Institute of Vibrational Field Dynamics.